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Liver Injury and Failure in Critical Illness

Thomas Horvatits,[1,2] Andreas Drolz,[1,2] Michael Trauner,[2] and Valentin Fuhrmann[1-3]

The frequency of acquired liver injury and failure in critical illness has been significantly increasing over recent decades. Currently, liver injury and failure are observed in up to 20% of patients in intensive care units and are associated with significantly increased morbidity and mortality. Secondary forms of liver injury in critical illness are divided primarily into cholestatic, hypoxic, or mixed forms. Therefore, sufficient knowledge of underlying alterations (e.g., hemodynamic, inflammatory, or drug induced) is key to a better understanding of clinical manifestations, prognostic implications, as well as diagnostic and therapeutic options of acquired liver injury and failure. This review provides a structured approach for the evaluation and treatment of acquired liver injury and failure in critically ill patients. (Hepatology 2019;70:2204-2215).

Hepatic failure is a frequently seen type of organ failure in critically ill patients.[1] Liver injury and failure contribute to dramatically increased morbidity and mortality.[1-3] Management of liver injury in critical illness represents a major multidisciplinary challenge. Liver injury in critical illness may be divided into groups: On the one hand, liver cirrhosis and subsequent acute-on-chronic liver failure (ACLF) and primary acute liver failure (ALF) are well-known entities of primary hepatic dysfunction.[4-8] However, secondary or acquired liver injury and failure (primarily without underlying liver disease) such as hypoxic liver injury (HLI), cholestatic dysfunction in critical illness, sclerosing cholangitis in critical illness (SC-CIP), and drug-induced liver injury (DILI) represent by far the most frequent forms of hepatic injury in critical illness. Although liver failure in the absence of chronic liver diseases refers primarily to the unique entity of ALF, defined as onset of hepatic encephalopathy and hepatic coagulopathy in patients without underlying liver disease (despite some exceptions), secondary forms of liver injury can also fulfill diagnostic criteria of ALF and are consequently incorporated into the recent European guidelines on ALF.[4,6] In contrast to primary ALF, liver transplantation is usually contraindicated in patients with secondary ALF.

Secondary forms of liver injury and failure are striking predictors of worse outcome and prolonged hospitalization.[9-11] Multiple factors are involved in the pathogenesis of hepatic dysfunction in critical illness. An overview of typical acquired liver injuries and their epidemiology in critical illness is found in Fig. 1 and Table 1, respectively.

Abbreviations: AKI, acute kidney injury; ALF, acute liver failure; AP, alkaline phosphatase; BA, bile acid; CCL2, chemokines; CO, cardiac output; COPD, chronic obstructive pulmonary disease; DAMPS, damage-associated molecular patterns; DILI, drug-induced liver injury; ERC, endoscopic retrograde cholangiography; GGT, gamma-glutamyl transpeptidase; HLI, hypoxic liver injury; ICU, intensive care unit; IL-1β, interleukin-1β; IL-6, interleukin-6; iNOS, inducible nitric oxide synthase; INR, international normalized ratio; NAC, N-acetylcysteine; NFkB, nuclear factor-kappa B; norUDCA, norursodeoxycholic acid; PAMPs, pathogen-associated molecular patterns; PaO_2, partial pressure of oxygen; PI3Kγ, phosphatidylinositol-3–kinase gamma; RRT, renal replacement therapy; SC-CIP, sclerosing cholangitis in critically ill patients; and UDCA, ursodeoxycholic acid.

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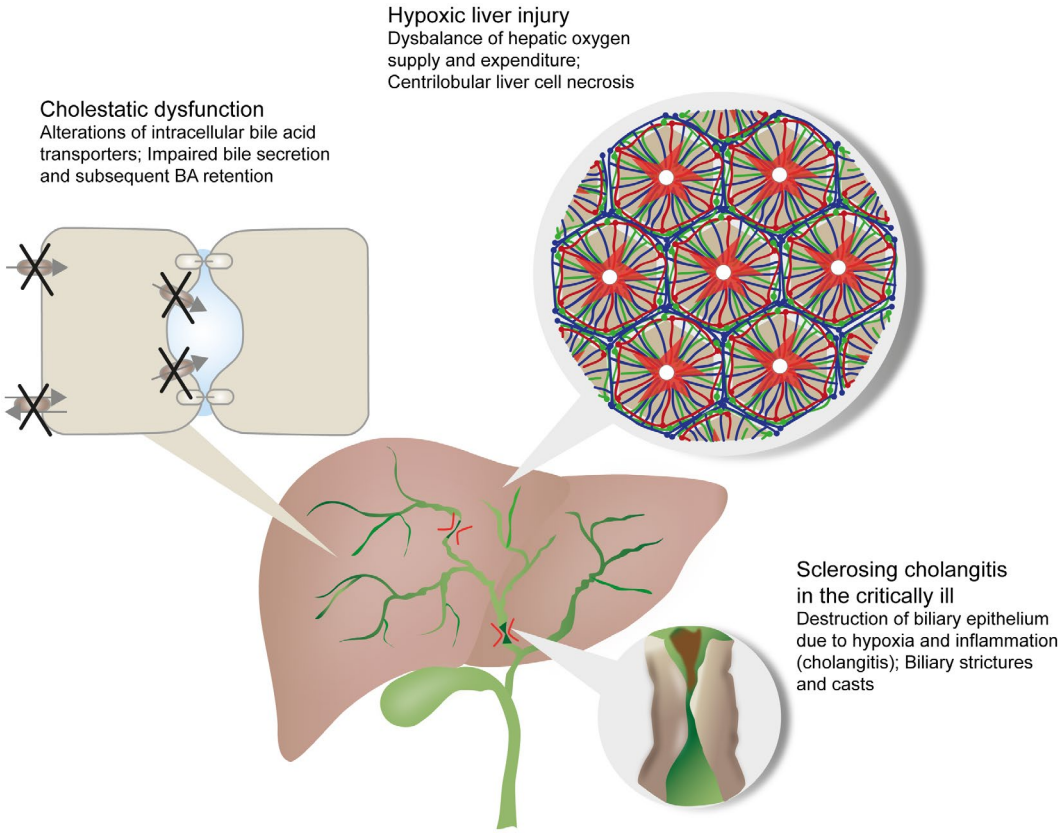


FIG. 1. Typical hepatic injuries in critical illness.

Jaundice and Cholestatic Dysfunction

DEFINITION AND EPIDEMIOLOGY

Cholestasis in critically ill patients is generally characterized by impaired bile formation and bile acid (BA) excretion.[12] Intrahepatic cholestasis is a consequence of alterations in signaling and transport at the hepatocellular level, while extrahepatic cholestasis is defined as the inability of a sufficient bile excretion due to mechanical bile duct obstruction. Currently, there is no uniform definition of cholestasis in critically ill patients. According to most studies, total bilirubin levels greater than 2 mg/dL represent a pragmatic cutoff in daily practice.[9,13-15]

ARTICLE INFORMATION:

From the [1]Department of Intensive Care Medicine, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; [2]Division of Gastroenterology & Hepatology, Department Internal Medicine 3, Medical University of Vienna, Vienna, Austria; [3]Department of Medicine B, Gastroenterology and Hepatology, University Münster, Münster, Germany.

ADDRESS CORRESPONDENCE AND REPRINT REQUESTS TO:

Valentin Fuhrmann, Prof.Dr.med.
Department of Medicine B for Gastroenterology and Hepatology
University Hospital Muenster
Albert-Schweitzer-Campus 1
48149 Muenster, Germany
E-mail: vfuhrmann@outlook.de
Tel.: +0049-251-83-57935

TABLE 1. Prevalence and Outcome of Different Liver Disease Entities at the ICU

	Prevalence	Hospital Mortality	Evaluation for Liver Transplantation
Decompensated cirrhosis/ACLF*[5,7,77,84,85]	1.6%-5.4%	36%-86%	Yes
Acute liver failure[87-89]	<10 cases per million persons/year	23-53	Yes
Hypoxic liver injury[2,61,64,65,67]	1%-15%	45%-57%	No
Cholestatic liver injury[3,9,15,37]	11%-36%	27%-48%	No†

*Cirrhosis as reason for admission.
†Possible in SSC-CIP.

Others use 2-fold to 3-fold elevation of alkaline phosphatase (AP) and/or gamma-glutamyl transpeptidase (GGT) to define cholestasis.[16,17]

Up to 20% of patients develop cholestasis during their intensive care unit (ICU) stay.[2,9,15] However, cholestatic alterations are more frequently found in septic shock (33%) than in cardiogenic shock (21%) or following surgery (5%), as indicated in Supporting Table S1.[18]

PATHOPHYSIOLOGY OF CHOLESTASIS IN THE CRITICALLY ILL

The clinical association of cholestasis and inflammation is well-known.[16] Recent insights demonstrate that alterations of hepatic transport and metabolism already occur in the first hours of critical illness.[18-20] Apart from infections with gram-negative bacteria, malperfusion, retention of toxic molecules, and pathogen-associated molecular patterns contribute to cholestasis in critical illness through inflammation-mediated repression of hepatobiliary transporters and reduced canalicular contractility, resulting in impaired bile flow and BA retention.[21,22] BAs also have important metabolic signaling functions that regulate energy, lipid, and glucose homeostasis.[23] BAs have both pro-inflammatory and anti-inflammatory properties, probably depending on serum concentrations and state of critical illness.[17,18] It is under debate whether cholestasis and BA retention reflect a distinct pathophysiological condition, an expression of severity of organ failure or instead a compensatory mechanism with potential beneficial actions (e.g., through inhibitions of hepatic inducible nitric oxide synthase [iNOS] activity) (Fig. 2).[17,24]

Recently, it has been shown that neonatal hyperbilirubinemia may protect newborns from *Streptococcus*

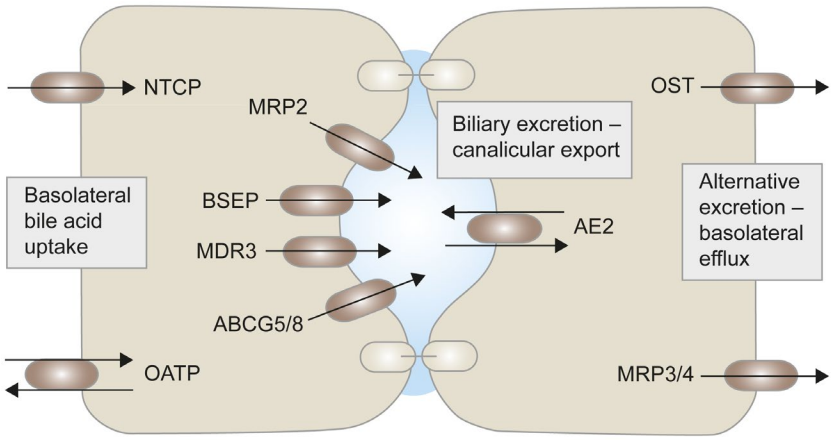


FIG. 2. Bile acid metabolism and biliary transporters in the hepatocyte. Abbreviations: AE2, anion exchanger 2; BSEP, bile salt export pump; MDR3, multidrug resistance gene 2; MRP, multidrug resistance–associated protein; NTCP, sodium-taurocholate cotransporting polypeptide; OATP, organic anion transporting polypeptide; and OST, organic solute transporter alpha.

agalactiae associated sepsis, thereby suggesting an evolutionary role of hyperbilirubinemia.[25]

Heme degradation products Z-BOX A and B, which are elevated in cholestatic liver injury, show dose-dependent intrahepatic and extrahepatic cytotoxicity.[26] Collectively, impaired hepatobiliary transport and BA signaling contributes to cholestasis and may have widespread implications for control of metabolism, inflammation, gut integrity, and microbiota. Detailed information on pathophysiology of cholestatic dysfunction and impact of serum bile acids is shown in the Supporting Information.

DIAGNOSIS

Serum total bilirubin, AP, and/or GGT are surrogate parameters for cholestatic impairment. Although the complexity of cholestatic dysfunction is not sufficiently characterized by serum bilirubin,[17,18] isolated elevation of conjugated bilirubin may reflect impaired biliary secretion as early sign of cholestasis of sepsis.[21] Several circumstances influence bilirubin metabolism in critical illness. Hemolysis results in unconjugated hyperbilirubinemia, whereas hepatic dysfunction *per se* as well as extrahepatic occlusion result primarily in conjugated or in mixed conjugated and unconjugated hyperbilirubinemia. AP and GGT, sensitive but unspecific indicators for cholestasis, are located primarily in hepatocytes and cholangiocytes but also in kidney or bone tissue. However, GGT activity is increased in several forms of liver and bile duct disorders, ranging from heavy drinkers, hepatitis, cirrhosis, up to infections or sclerosing cholangitis as well as primary or secondary neoplasms and is associated with all-cause mortality.[27] Incidence of cholestatic dysfunction depends on the definition used and ranges between 13% (defined as AP > 400 IU/L and GGT > 80 IU/L) and 17% (defined as bilirubin > 3 mg/dL).[15]

Serum BAs may indicate changes in biliary formation and subsequent cholestatic impairment earlier and with higher prognostic accuracy than bilirubin.[12,16,18] Total BAs potentially represent a feasible biomarker for severity of disease, as they predict short-time mortality independently of serum bilirubin and severity of underlying disease as reported in a recent study.[18] MicroRNAs, central regulators of posttranscription gene expression, may serve as potential biomarkers for several liver diseases.[28] However, their relevance in ICU-related liver injury remains to be determined.

In patients with ongoing cholestatic impairment, imaging should be considered. Jaundice due to biliary obstruction (e.g., cholelithiasis/calculous cholecystitis) can be diagnosed by bedside abdominal sonography. In case of extrahepatic cholestasis (common bile duct > 7 mm in ultrasound),[29] patients can then be selected for either computed tomography scan or magnetic resonance imaging/magnetic resonance cholangiopancreatography (diagnostic accuracy for detection of benign/malignant diseases: 83%/91% and 98%/98%, respectively).[30] Endoscopic retrograde cholangiography (ERC) may help to rule out SC-CIP (see subsequently) in unclear prolonged jaundice. Nevertheless, intrahospital transport of critically ill patients may be associated with adverse events. Use of checklists for intrahospital transport (guiding physicians and nurses) can enhance patients' safety.[31]

Elevated liver stiffness (>18 kPa) assessed by transient elastography may predict early liver dysfunction and outcome in critical illness.[32] However, transient elastography could not be adequately performed in more than 30% of cases.[32] A proposed diagnostic approach for cholestatic dysfunction is shown in Supporting Fig. S2.

CLINICAL PRESENTATION AND COURSE OF DISEASE

Inflammation-associated cholestasis occurs in the context of intra-abdominal and extra-abdominal infections, sepsis, and septic shock.[16,33] Typical risk factors for new onset of cholestasis are preexisting liver disease, shock, severe respiratory insufficiency, renal failure, as well as total parenteral nutrition.[21,34] Vasopressors decrease bile flow and excretion of BAs possibly through intracellular increase of calcium in hepatocytes.[35] Furthermore, bone marrow transplantation, major surgery, as well as severe trauma may contribute to jaundice and cholestatic dysfunction.[34,36,37] Up to 66% of patients had biliary sludge in abdominal ultrasound 5 days after ICU admission.[15]

Cholestasis and jaundice are associated with critical illness. Cholemic nephropathy, a histopathological finding in the context of longstanding jaundice, leads to epithelial cell injury and intratubular cast formation and subsequent renal failure.[38]

PROGNOSIS

Cholestatic dysfunction in critically ill patients is associated with more than 2-fold increased mortality rates.[3,9,37] Early hepatic dysfunction had a strong impact on survival in critically ill patients without prior liver disease. Patients with serum bilirubin levels between 2 and 3 mg/dL had a 27% hospital mortality rate, in patients with bilirubin 6-10 mg/dL and in those with bilirubin greater than 10 mg/dL, the hospital mortality was 38% and 48%, respectively.[9] New onset of jaundice was associated with increased rates of renal failure, new onset of infections, and gastrointestinal complications. Furthermore, 1-year survival in patients with HLI was dramatically decreased to 8%.[3] Increased serum BA was significantly associated with elevated 28-day mortality in critically ill patients independently of bilirubin levels and severity of underlying disease.[18,19]

THERAPEUTIC ASPECTS

Currently, there is no established targeted treatment of cholestasis of critically ill patients. Therapy should focus on prompt removal of the underlying condition causing hepatic impairment. Current guidelines recommend hemodynamic stabilization, optimizing perfusion and adequate treatment of infections (including targeted initiation of broad-spectrum antibiotic treatment and subsequent adaption of antimicrobial therapy to culture results).[39]

Glucose control, particularly intensive insulin therapy, reduces the rate of cholestatic dysfunction and biliary sludge in critically ill patients.[15,40] Early enteral nutrition is generally recommended as standard of care in critical illness.[41] It should only be postponed in uncontrolled shock, severe acidosis, hyperlactatemia and/or hypoxemia, increased gastric residual volume, abdominal obstruction, and/or abdominal compartment syndrome.[42] Details are found in Supplementary Table S2.

Resins and BA sequestrants can lower serum BA levels by preventing ileal reabsorption and removal from enterohepatic circulation. To date, several farnesoid-X-receptor ligands are available that directly intervene with BA homeostasis and gut integrity with inflammatory properties, but have not yet been tested in cholestasis of sepsis and critical illness. Ursodeoxycholic acid (UDCA) has been shown to protect cholangiocytes against cytotoxic acids and to lower concentration of hydrophobic BAs and is frequently used in cholestatic liver disease.[43,44] A pilot study (presented as abstract) showed biochemical improvement in critically ill patients receiving high-dose UDCA, although it did not affect outcome.[45] Norursodeoxycholic acid (norUDCA) is a novel side-chain-shortened derivative of UDCA. It is a promising substance in the treatment of cholestatic impairment and bile duct diseases due to its beneficial effects on cholestasis and counteraction to fibrosis, steatosis, and hepatic inflammation.[46] To date, clinical evidence pertains primarily to sclerosing cholangitis,[46] but not to critical illness. Randomized controlled trials are needed to assess the role of UDCA and norUDCA in critically ill patients.

Renal failure and subsequent need for dialysis was reported to be significantly higher in patients with jaundice than in patients without jaundice.[3] Advanced extracorporeal systems such as albumin dialysis (i.e., Molecular Adsorbents Recirculatory System as well as Fractionated Plasma Separation and Adsorption) are able to remove albumin bound substances like BAs and water-soluble substances. Randomized, controlled studies demonstrated that albumin dialysis devices improve hepatic encephalopathy and systemic hemodynamics significantly and reduce BAs, although they did not demonstrate improved survival.[47-51] Future studies should compare different dialysis devices in critically ill patients with liver injury and failure.

CHOLANGIOPATHY IN CRITICAL ILLNESS

Cholestasis induced by critical illness as a result of functional alterations at the hepatocellular and canalicular level is usually reversible. However, a small proportion of patients develop irreversible biliary and hepatic damage due to progressive sclerosing cholangitis. SC-CIP is a rare but clinically relevant complication described in patients with severe trauma, burn injury, severe respiratory, or hemodynamic failure. A study among 2,633 patients who were listed or underwent liver transplantation identified 16 cases of SC-CIP (0.6%).[52] Malperfusion, hypoxia, and recurrent inflammatory stimuli due to episodes of bacterial cholangitis appear to be the main trigger for destruction of the biliary epithelium.[53] The gold standard for diagnosis of SC-CIP is ERC.[54,55] Typical endoscopic

findings are the presence of biliary casts, multiple irregular strictures, and rarefication of the biliary tree that can be treated during ERC immediately.[56] Antimicrobial treatment covering typical pathogens of biliary tract infections should be administered according to local microbiological circumstances. SC-CIP is frequently a progressive disease that potentially results in liver cirrhosis with subsequent need for liver transplantation.[57] However, only a minority of patients with SC-CIP is clinically eligible for transplantation. Continuous inflammatory damage of the biliary system impairs prognosis in nontransplant candidates (median survival of 13 months).[53]

DRUG-INDUCED LIVER INJURY IN CRITICAL ILLNESS

Data on the incidence of DILI in the ICU are lacking. Many drugs are potentially capable of causing DILI, and typical patterns and latencies are shown in Supporting Fig. S2. One of the most common drugs causing intrinsic (usually dose-dependent) DILI is acetaminophen.[58-61] In the ICU setting, typical and more frequent causes of elevated liver function parameters (such as HLI and cholestatic dysfunction of critical illness) should be excluded.[62-64] The hallmark of DILI therapy is the withdrawal of the suspected causal drug. More information on pathogenesis and management of DILI is found in the Supporting Information.

Hypoxic Liver Injury

The massive rise of aminotransferase levels in combination with increasing international normalized ratio (INR) in critically ill patients is an impressive and challenging setting. HLI is the most common cause of massive elevation of transaminase levels in critically ill patients, but also (in particular acetaminophen associated) DILI has to be considered.[2,6,58,62] The typical histopathological pattern in HLI is centrilobular liver cell necrosis. This clinical syndrome has been labeled by various terms over time: "passive congestion of the liver," "shock liver," "ischemic hepatitis," and "hypoxic hepatitis." The changing labeling illustrates the evolution of the pathophysiological understanding of the contributing factors leading to centrilobular liver cell necrosis. At present, the syndrome is primarily referred to as "hypoxic liver injury," due to the multitude of factors that can lead to centrilobular liver cell necrosis. As a result of its clinical significance in everyday hospital life, the European Association of Studies of the Liver included HLI as a main cause of secondary ALF in their Clinical Practical Guidelines on the management of ALF.[6]

DEFINITION AND EPIDEMIOLOGY

HLI is defined as a massive rise of aminotransferase levels in a setting of cardiac, circulatory, or respiratory failure. Although other putative causes should be excluded, patients in ICU with massive rise of aminotransferase levels (many studies used a cutoff of ≥ 20 the upper limit of normal) should be considered having HLI until proven otherwise.[2,3,10,62,65,66]

HLI is by far the most common cause of massive elevated aminotransferase levels in hospitalized patients.[67] The prevalence of HLI ranges between 1% and 12% in ICU and may be even higher in patients with cardiogenic shock, in whom the prevalence was reported to be greater than 20%.[62,65,66,68,69]

Acute heart failure and cardiogenic shock are present in 50%-70% of patients with HLI, whereas sepsis is observed in 13%-32% and respiratory failure in 10%-20%.[62,65,68,70] Many patients who develop HLI suffer from chronic diseases. Cardiomyopathy is present in up to 40% of the patients, chronic obstructive pulmonary disease in 20%, and cirrhosis was reported in 15% of patients in an ICU series.[62] In patients with cirrhosis and HLI, portal vein thrombosis was frequently observed.[71] Importantly, in more than 75% of the cases, HLI onset appears to be a multifactorial syndrome in which more than one of the underlying causes contribute to centrilobular liver cell necrosis.[62]

PATHOPHYSIOLOGY OF HLI

The hepatic blood flow accounts for approximately 20%-25% of the total blood flow. The dual blood supply of the liver through the portal vein and hepatic artery is a unique feature of the hepatic vasculature. In a stable condition, the portal vein contributes to 75%-80% of the livers' blood supply, and the remaining 20%-25% are provided by the hepatic artery. Due to this dual blood supply, the liver can compensate

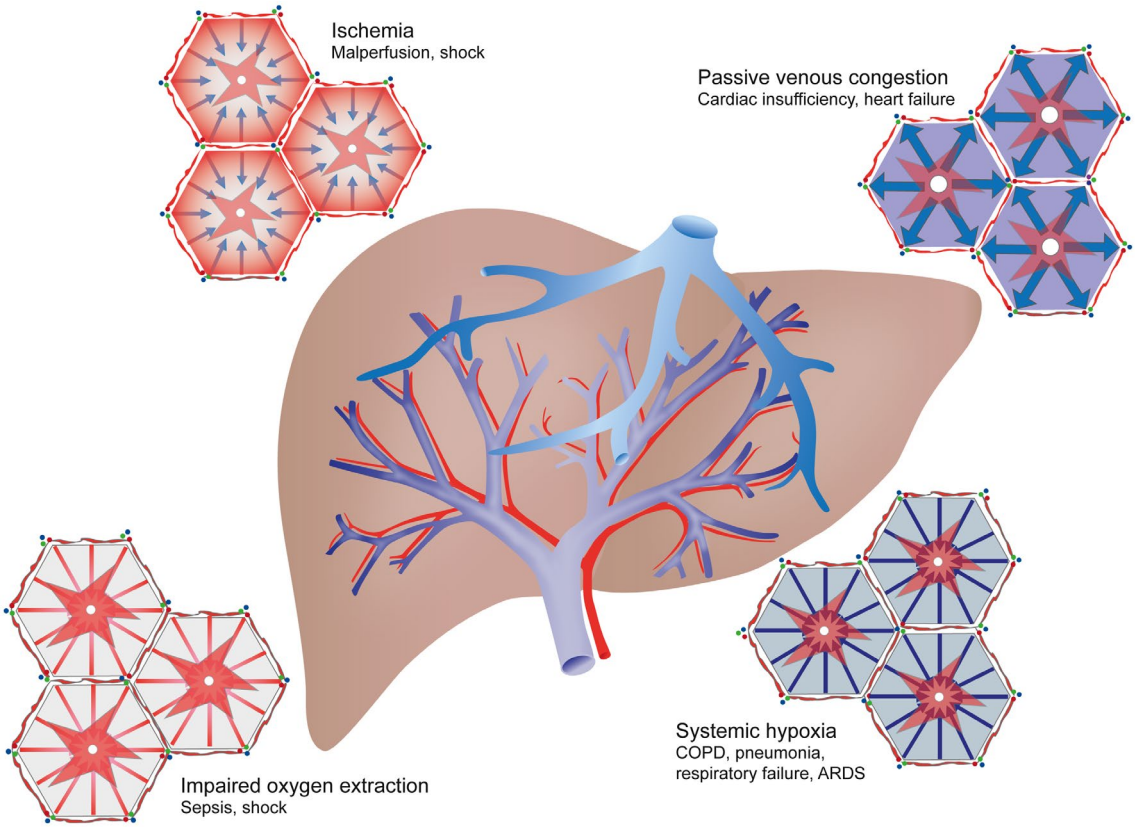


FIG. 3. Pathophysiological alterations in hypoxic liver injury. Abbreviations: ARDS, acute respiratory distress syndrome; COPD, chronic obstructive pulmonary disease.

for most of the hemodynamic alterations and hits. This is carried out by several mechanisms like the hepatic arterial buffer response (HABR).[72] HABR represents the ability of the hepatic artery to compensate for changes in the portal vein flow. Apart from the intrinsic arterial autoregulation, the hepatic artery dilates following decreased portal blood flow and *vice versa*. This adaptive mechanism relies primarily on adenosine. In the case of reduced blood flow, accumulation of adenosine leads to dilatation of the hepatic arterioles.[72] An increase in hepatic arterial blood flow is able to buffer 25%-60% of the decreased portal flow. Furthermore, hepatic oxygen extraction can be increased up to 90% in the case of reduced hepatic perfusion and venous congestion, due to the high permeability of hepatic sinusoids that promote diffusion of oxygen to hepatic cells.[73] Liver sinusoidal endothelial cells respond to inflammatory signals like endotoxin by an iNOS-dependent endothelial dysfunction contributing to a decreased vasodilatory response and decreased endothelial nitric oxide synthase phosphorylation.[74] An overview of different pathomechanisms contributing to HLI is found in Fig. 3 and in the Supporting Information.

DIAGNOSIS

Aminotransferase levels as marker for hepatocellular injury in conjunction with the clinical setting of acute cardio-circulatory, respiratory, or hemodynamic injury are the diagnostic cornerstone of HLI. Aminotransferase levels rise with a delay of several hours following the acute hit, and aspartate aminotransferase (AST) levels are in the initial phase higher than alanine aminotransferase (ALT) levels. Aminotransferase levels start to decline within several hours up to 1 day after successful treatment of the underlying problem and consecutive stabilization of the patient (AST levels decline faster than ALT levels). Recently, it could be demonstrated that the mechanism of injury during HLI includes

mitochondrial damage and DNA fragmentation.[75] In addition to conventional laboratory assessment, indocyanine green plasma disappearance rate is an accurate marker of hepatic function with prognostic properties in HLI.[76] Imaging techniques of the liver may provide additional information like dilated hepatic veins following congestion or reduced perfusion of the hepatic artery or portal vein. However, the most important imaging technique in patients with HLI is transthoracic echocardiography, as it provides important data on cardiac function and impairment, which is by far the most common underlying cause of HLI.

Liver biopsy is usually not warranted due to the typical clinical pattern of HLI and the potential of procedural complications. Supporting Fig. S3 illustrates a proposed diagnostic approach for HLI.

CLINICAL PRESENTATION AND COURSE OF DISEASE

Patients with HLI are usually of advanced age. The ratio of men to women is 2:1. Vasopressor therapy is necessary in 50%-85%. Furthermore, 70%-80% of the patients require mechanical ventilation, and renal replacement therapy (RRT) is performed in up to 41%.[66] More than 80% of the patients develop HLI within the first 24 hours after ICU admission. Several prospective studies were able to demonstrate that the liver *per se* is a trigger of multiorgan failure in HLI.[3,76-78] Hepatopulmonary syndrome (HPS) is observed in more than 40% of patients with HLI and was associated with a significant impaired oxygenation within the first 48 hours after diagnosis. Complete resolution of HPS was observed following normalization of the liver function during follow-up.[77] New onset of jaundice was observed in 36% of HLI patients within 2-6 days after HLI, and was associated with dramatically increased mortality rates.[3] Patients with jaundice following HLI are prone to new onset of clinically relevant complications like infections, acute kidney injury (AKI), and gastrointestinal injuries. Twenty five percent of the patients present signs of hepatic encephalopathy following HLI and fulfilled criteria of ALF.[6,78] Arterial ammonia levels may provide additional information in the risk assessment of patients with HLI.[78] Spontaneous hypoglycemia was observed in more than 10% of patients with HLI.[62] AKI is found in more than 80% of patients with HLI. Although the causal relationship of AKI and HLI is frequently difficult to establish due to the multitude of factors that contribute to AKI, its clinical relevance is apparent: More than 50% of patients with HLI have AKIN (AKI network) stage 3 and many of them require RRT.[66] In addition, induction and aggravation of systemic inflammatory processes and coagulation disorders may be aggravated following HLI. Figure 4 illustrates typical interactions of the liver as triggers of organ injury and failure.

PROGNOSIS

Critically ill patients with HLI have significantly decreased survival rates. Twenty-eight-day mortality ranges between 40% and 60%, and 1-year mortality rate is about 75%.[62,65,68,70,79] The highest mortality rates are observed in patients with septic shock, high severity of organ failure (i.e., Sequential Organ Failure Assessment score ≥ 11), and patients with preexistent chronic liver diseases.[62] Apart from the type and severity of the underlying disease causing HLI, the degree of hepatic impairment illustrated by INR was an independent predictor of mortality in two large studies.[62,70] In addition, duration of HLI has a prognostic impact. Increased mortality in patients with prolonged hepatic injury reflects the impossibility to reverse the underlying HLI-causing disease.[62] This prolonged persistence of HLI triggers progression of organ failure *per se*.

THERAPEUTIC OPTIONS

Early recognition of HLI and subsequent therapy of the underlying diseases contributing to onset HLI is of central importance. Liver transplantation is not indicated in patients with HLI. For details on clinical management, see Supporting Table S2.

In experimental studies, statins (inhibitors of 3-hydroxy-3-methylglutaryl-conenzyme A reductase) improved hepatic microcirculation and endothelial function, dampened the inflammatory response, and reduced ischemia/reperfusion injuries.[74,80,81] Accordingly, pretreatment with statins was associated with reduced rates of HLI in a prospective cohort study of critically ill patients.[82] Spontaneous hypoglycemia is associated with excessive high mortality rates in HLI and should be treated immediately.[62] Although hyperammonemia is frequently observed in patients with HLI, the impact of ammonia-lowering

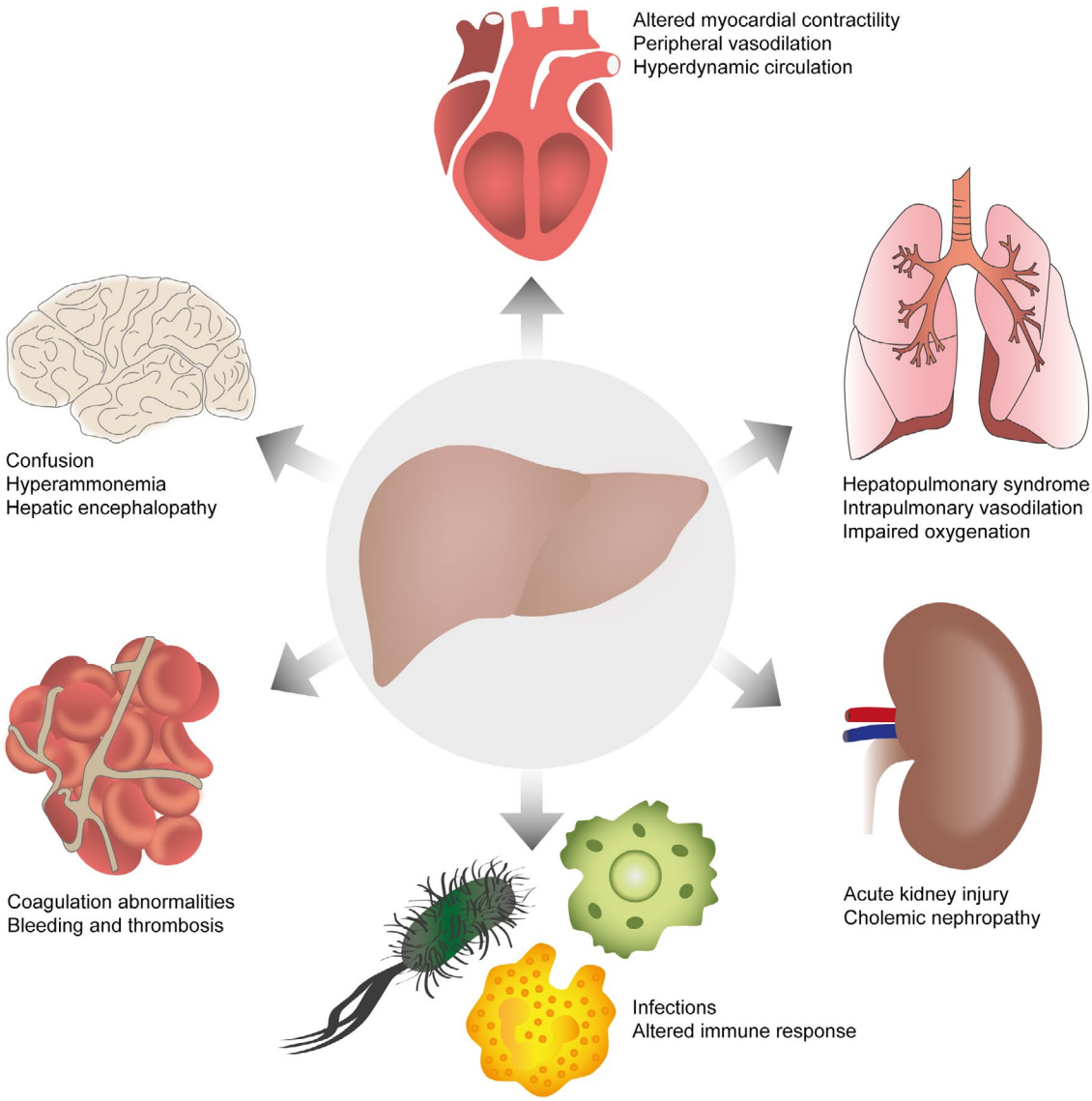


FIG. 4. Liver as a trigger of organ injury and failure.

measures has not been assessed in this patient population so far. Initiation of RRT improved 28-day survival in a large observational analysis of patients with HLI.[66] Although mortality was high in patients with HLI and RRT, kidney function recovered in 71% of the ICU-surviving patients with 1-year survival rates of approximately 60%. Advanced dialysis devices, which are capable of removing albumin-bound substances in addition to water-soluble molecules, may improve systemic and splanchnic hemodynamics in patients with HLI, ALF, and ACLF.[47,48,83] A randomized, controlled study observed a trend of better outcome using albumin dialysis in patients with cholestatic HLI.[84] However, future studies are needed to assess the superiority of one dialysis device to others in these patients.

In conclusion, liver injury and failure are common clinical problems affecting more than 20% of critically ill patients. Cholestasis and HLI are the most frequent clinical presentations. Early diagnosis of hepatic injury, identification of underlying hepatic and extrahepatic problems, and early initiation of primarily symptomatic therapy are the cornerstones of clinical management.

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Author names in bold designate shared co-first authorship.

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